WORKED SOLUTION

STEP II 1994, Question 6

Pure • Telescoping Trigonometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q6

The key idea

Everything rests on one small identity: $\tan x = \cot x - 2\cot 2x$. It turns each term of the sum into a difference of two consecutive terms of the form $\frac{1}{2^r}\cot\frac{\theta}{2^r}$, so the sum telescopes, and that same identity is exactly what makes the inductive step work. The infinite sum then needs only the limit $\frac{\sin\varphi}{\varphi}\to 1$.

Solution

A useful identity

For any x with $\sin x$, $\cos x$ and $\sin 2x$ all non-zero,

$$\cot x - \tan x = \frac{\cos x}{\sin x} - \frac{\sin x}{\cos x} = \frac{\cos^2 x - \sin^2 x}{\sin x\cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x,$$

so

$$\tan x = \cot x - 2\cot 2x. \qquad (*)$$

We shall only use $(*)$ with $x = \theta/2^r$ for $r \geq 1$. Since $0 < \theta < \pi$, such an x lies in $(0, \frac{\pi}{2})$, so $\sin x$, $\cos x$ and $\sin 2x$ are all positive and every tan and cot we write down is defined.

The finite sum, by induction

Write S_n for the left-hand side, $\sum_{r=1}^{n}\frac{1}{2^r}\tan\frac{\theta}{2^r}$, and let $P(n)$ be the statement

$$S_n = \frac{1}{2^n}\cot\frac{\theta}{2^n} - \cot\theta.$$

Base case $n = 1$. Put $x = \theta/2$ in $(*)$ and halve:

$$S_1 = \tfrac{1}{2}\tan\tfrac{\theta}{2} = \tfrac{1}{2}\cot\tfrac{\theta}{2} - \cot\theta,$$

which is $P(1)$.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$. Put $x = \theta/2^{k+1}$ in $(*)$, so that $2x = \theta/2^k$, and divide by 2^{k+1}:

$$\frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}} = \frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k}.$$

Adding this to $P(k)$,

$$\begin{aligned} S_{k+1} &= S_k + \frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}} \\ &= \left(\frac{1}{2^k} \cot \frac{\theta}{2^k} - \cot \theta \right) + \left(\frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k} \right) \\ &= \frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \cot \theta, \end{aligned}$$

which is $P(k+1)$. By induction $P(n)$ holds for every $n \geq 1$ and every $\theta \in (0, \pi)$, as required.

Ask yourself How would you spot the identity $(*)$ if you hadn't seen it before?

Work backwards from the induction. Going from $n = k$ to $n = k+1$, the right-hand side changes by $\frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k}$, and the left-hand side by $\frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}}$. Multiplying by 2^{k+1} and writing $x = \theta/2^{k+1}$, the two changes agree precisely when $\tan x = \cot x - 2 \cot 2x$. So the question tells you which identity to prove; you then check it with the double-angle formulae.

The infinite sum

The terms are all positive, and we have an exact formula for the partial sums, so it suffices to find the limit of S_n as $n \to \infty$. Let $\varphi_n = \theta/2^n$, so $\varphi_n \to 0^+$ and $\frac{1}{2^n} = \frac{\varphi_n}{\theta}$. Then

$$\frac{1}{2^n} \cot \frac{\theta}{2^n} = \frac{\varphi_n}{\theta} \cdot \frac{\cos \varphi_n}{\sin \varphi_n} = \frac{1}{\theta} \cdot \frac{\varphi_n}{\sin \varphi_n} \cdot \cos \varphi_n \longrightarrow \frac{1}{\theta} \cdot 1 \cdot 1 = \frac{1}{\theta},$$

using the standard limit $\frac{\sin \varphi}{\varphi} \to 1$ as $\varphi \to 0$ and $\cos \varphi \to 1$. Since $\cot \theta$ does not depend on n,

$$\sum_{r=1}^{\infty} \frac{1}{2^r} \tan \frac{\theta}{2^r} = \lim_{n \to \infty} S_n = \frac{1}{\theta} - \cot \theta,$$

which is the required result. In particular the series converges for every $\theta \in (0, \pi)$.

Ask yourself Why is it legitimate to say "the infinite sum equals the limit of S_n"?

That is the definition of the sum of an infinite series: $\sum_{r=1}^{\infty} a_r$ means $\lim_{n \to \infty} \sum_{r=1}^{n} a_r$, when that limit exists. We have shown the limit exists by computing it, so nothing more is needed.

Checks

- $\theta = \frac{\pi}{2}$: the right-hand side is $\frac{2}{\pi} - 0 \approx 0.63662$. Summing the first 60 terms of the series numerically also gives 0.6366198.
- $\theta = 1$: both the series and $1 - \cot 1$ give 0.3579074.
- Small θ: $\tan \frac{\theta}{2^r} \approx \frac{\theta}{2^r}$, so the series is about $\theta \sum 4^{-r} = \frac{\theta}{3}$. This agrees with the expansion $\frac{1}{\theta} - \cot\theta = \frac{\theta}{3} + O(\theta^3)$.
- As $\theta \to \pi^-$, $\cot\theta \to -\infty$ and the first term $\frac{1}{2}\tan\frac{\theta}{2} \to +\infty$, so both sides blow up together, as they should.

Common slips

- Getting the factor of 2 in $(*)$ wrong, for example writing $\tan x = \cot x - \cot 2x$. Test it at $x = \frac{\pi}{4}$: $1 = 1 - 2 \cdot 0$.
- Claiming $\frac{1}{2^n}\cot\frac{\theta}{2^n} \to 0$ because $\frac{1}{2^n} \to 0$. The cotangent grows like $2^n/\theta$, so the product tends to $\frac{1}{\theta}$, not 0.
- Forgetting to say why the tangents and cotangents are defined: it is the condition $0 < \theta < \pi$ that keeps every angle $\theta/2^r$ inside $(0, \frac{\pi}{2})$.

Beyond STEP

The function $\frac{1}{\theta} - \cot\theta$ belongs to a famous family: differentiating the partial-fraction expansion $\cot\theta = \frac{1}{\theta} + \sum_{k\geq 1} \frac{2\theta}{\theta^2 - k^2\pi^2}$ gives the Basel sum $\sum 1/k^2 = \pi^2/6$, and Euler obtained such results from the infinite product for $\sin\theta$. Taking logarithms and differentiating in Viète's product $\frac{\sin\theta}{\theta} = \prod_{r\geq 1} \cos\frac{\theta}{2^r}$ gives exactly this question's series, which is another way to see the result. Making such manipulations of infinite sums and products rigorous is a first-year analysis topic; see Abbott, *Understanding Analysis*.

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